

Question number	Answer	Additional guidance	Mark
5(a)	choose an item: • B is the correct answer A is not the correct answer as the reduction of carbon does not produce water C is not the correct answer as the phosphorylation of ADP does not produce water D is not the correct answer as the phosphorylation of glucose does not produce water		(1)

Question number	Answer	Additional guidance	Mark
5(b)(i)	Choose an item: B is the correct answer A is not the correct answer as 1.3 is not the rate of carbon dioxide production at 60 mins C is not the correct answer as 2.5 is not the rate of carbon dioxide production at 60 mins D is not the correct answer as 3.08 is not the rate of carbon dioxide production at 60 mins		(1)

Question number	Answer	Additional guidance	Mark
5(b)(ii)	An explanation that includes two of the following points: - the respiratory quotient decreased (during the exercise) (1) (switching to) {lipid/fat/triglyceride} being used (rather than carbohydrate)	accept suitably named compounds ignore ref to anaerobic respiration occurring? ignore ref to change in CO₂ production?	(2)

Question number	Answer	Additional guidance	Mark
5(c)(i)	A calculation showing the following steps: • calculation of total number of 80-year-olds (1) • calculation number of 80-year-olds who have cancer and are non-smokers with COPD and given in standard form (1)	(105500 + 158600=)264100 (264100 x 7.5/100=) 1.98 x 10⁴ should be a whole number	(2)
	2		
	1		
	1.9 x 10 ⁴ to 2.1 x 10 ⁴		
	19015 to 20600		
	264100		

Question number	Answer	Additional guidance	Mark
5(c)(ii)	An answer that includes four of the following points: • ventilation and heart rate increase (1) • to deliver sufficient oxygen to body cells for aerobic respiration (1) • (because damage to walls of alveoli/COPD) will reduce surface area for {diffusion/gas exchange} (1) • (thicker) mucus will increase diffusion distance (1) • (resulting in) reduced rate of oxygen diffusion into blood / lower oxygen content of blood} (1)	can piece together mp1 only accept descriptions of breathing rate and heart rate increasing accept anaerobic respiration occurs due to low oxygen accept to prevent anaerobic respiration occurring (due to insufficient oxygen) reduced elasticity of alveoli decreases concentration gradient for {diffusion / gas exchange} less oxygen entering into blood	(4)

